Testing Plan for Interactive Prototype 3

1. Project Background

This version of the prototype builds upon the previous immersive drawing system by introducing new physical and spatial interaction features to create a more realistic and responsive VR environment. It expands beyond drawing and erasing to include cabinet manipulation, lever-based scene control, and camera height adjustment through controller motion.

The new features include a **two-layer cabinet**—the first layer contains a pullable drawer with smooth sliding physics, while the second layer features a shutter door that rotates on a hinge. A **lever** attached to a vertical pole allows scene transitions: pulling it down to the maximum angle and holding for half a second switches to Scene 1, while pulling it upward to the opposite limit switches back to Scene 0. The **camera height control** enables users to hold the right controller's thumbstick and move their hand vertically along the Y-axis to adjust the camera height, simulating standing or crouching postures.

These additions aim to evaluate how multi-modal physical interactions affect user experience, realism, and spatial awareness in an XR creative workspace.

2. Testing Objectives

The main goal is to understand how intuitive, comfortable, and realistic these new interactions feel to users.
Specifically, the test will:

- Evaluate how natural it feels to open and close the drawer and shutter door.
- Assess the responsiveness and physical feedback of the lever for scene switching.

- Examine how intuitive and stable the camera height adjustment mechanism is.
- Gather participant feedback on usability, immersion, and control precision in a creative VR setting.

3. Testing Methodology

A mixed approach combining **think-aloud observation** and **task-based interaction testing** will be used. Participants will perform a set of structured tasks while verbalizing their thoughts. The researcher will observe and take notes on performance, comments, and behavioral cues.

Task Sequence

1. Open and close the **drawer** on the cabinet's first layer and try to put the pen in it, commenting on smoothness and realism.
2. Open and close the **shutter door** on the second layer using the hinge interaction.
3. Pull the **lever** to switch to another scene and push it back to return to the original.
4. Hold the **right controller thumbstick** and move the hand vertically to adjust the **camera height** until it feels natural.

Oral Inquiry

After completing all tasks, participants will be asked:

- Which cabinet interaction (drawer or shutter) felt more satisfying?
- Was the lever feedback clear when switching scenes?
- Did the camera height adjustment feel stable and natural?
- Which interaction felt most immersive or needed improvement?

4. Prototype Description

The prototype is built in **Unity (Built-in Render Pipeline)** and runs on a **Meta Quest headset** using Meta Building Blocks and XR Interaction Toolkit.
The cabinet objects use **colliders and joints** for realistic movement. The lever uses a **Configurable Joint** with limits and triggers scene transitions at ±70°. The camera control script tracks the right-hand controller's Y-axis movement while the thumbstick is pressed, updating the XR Rig's position dynamically. Scene transitions include fade effects for smooth switching.

5. Data Collection and Success Criteria

Data Collection

Both quantitative and qualitative data will be recorded:

- **Quantitative:** Task completion rate, number of errors, and time taken for each task.
- **Qualitative:** Participant comments on realism, comfort, intuitiveness, and immersion.
- **Ratings:** A 1–5 satisfaction scale for drawer, shutter, lever, and camera height interactions.

Success Criteria

- At least **80%** of participants complete all tasks without assistance.
- Average satisfaction rating of **4 or above** for cabinet and lever interactions.
- At least **70%** of participants find the camera height adjustment intuitive and convenient.
- Participants express improved spatial control and immersion compared to the previous prototype.

6. Testing Setup

Testing will be conducted in a safe, quiet environment suitable for VR movement. The setup includes:

- Unity build of the XR prototype running on **Meta Quest headset**
- Sufficient physical space for reaching and pulling motions
- Researcher observing and recording feedback
- Optional video capture or screenshots for later analysis

7. Testing Process

Each participant session lasts around **8–10 minutes**.

1. **Introduction (30 sec):** Explain the test purpose and emphasize that participants are evaluating the system, not themselves.
2. **Warm-up (1 min):** Demonstrate how to use the drawer, shutter, lever, and camera adjustment.
3. **Task performance (5–6 min):** Participants complete the five interaction tasks while thinking aloud.
4. **Debrief (2–3 min):** Researcher conducts a short discussion on comfort, preference, and perceived realism.

8. Expected Outcomes

The test is expected to provide insight into how environmental and camera-based interactions influence immersion and usability. Feedback will help refine physics calibration, interaction feedback, and comfort balance in future iterations. Ultimately, the goal is to achieve more natural, controllable, and enjoyable physical interaction experiences within VR creative environments.

9. Research Summary

Realistic physical interactions play a vital role in improving user immersion, spatial awareness, and perceived control in virtual environments. Lee et al. [1] emphasize that multi-modal hand-based interactions such as pulling, rotating, and pushing levers enhance realism and engagement compared to purely gestural input. This finding supports the inclusion of lever-based scene switching in the current prototype. Rothe et al. [2] further demonstrate that camera height and dynamic viewpoint adjustment directly affect users' sense of embodiment and comfort, validating the camera height control mechanism implemented through controller motion. Additionally, a comprehensive review of virtual reality interaction interfaces [3] outlines that physics-based mechanics—such as drawers, hinges, and levers—contribute to more intuitive and natural interactions by aligning virtual responses with users' real-world expectations.

Building on these studies, this prototype integrates drawer, shutter-door, lever, and camera-height interactions to explore how physical affordances influence immersion, usability, and realism in an XR creative workspace. The design aims to evaluate how embodied, physics-driven interactions improve control precision, spatial awareness, and user satisfaction compared to previous gesture-based systems.

10. References

[1] H. Lee, A. Restoles, and C. Vidal Jr., "A Virtual Reality Object Interaction System with Complex Hand Interactions," in Lecture Notes in Networks and Systems, 2023, pp. 91–103. Springer.

[2] S. Rothe, M. Busch, L. Nagel, and M. E. Latoschik, "The Impact of Camera Height in Cinematic Virtual Reality," in Proc. 24th ACM Symp. Virtual Reality Software and Technology (VRST), 2018.

[3] "A Survey on the Design of Virtual Reality Interaction Interfaces," Sensors, vol. 24, no. 19, p. 6204, 2024.